

ShikshaSetu — Technical Solution Report

Complete Architecture, Dual-Portal Workspaces & AI Permission Boundaries

- Next.js 14 App Router
- TypeScript
- SchoolGPT AI Core
- Supabase Database

ShikshaSetu — Complete Solution & Technical Implementation Documentation

Project Name: ShikshaSetu

Repository: `ankit4563564/ShikshaSetu`

Architecture: Next.js 14 App Router • TypeScript • TailwindCSS • Supabase • SchoolGPT AI Core

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1. Executive Summary

Traditional school management software (ERP) suffers from severe usability flaws: dense dashboards, cluttered tables, overwhelming metrics, and fragmented portals that force parents and teachers to navigate dozens of menus.

ShikshaSetu redefines school technology. Instead of treating school management as multiple disjointed products, ShikshaSetu is built as **ONE connected platform** with **TWO primary user experiences**:

1. **The Parent Experience**: Reassurance-first companion inspired by Apple Health, Google Maps, and Apple Wallet.
2. **The Teacher Experience**: Speed-optimized workstation that saves teachers up to 1.5 hours per week.

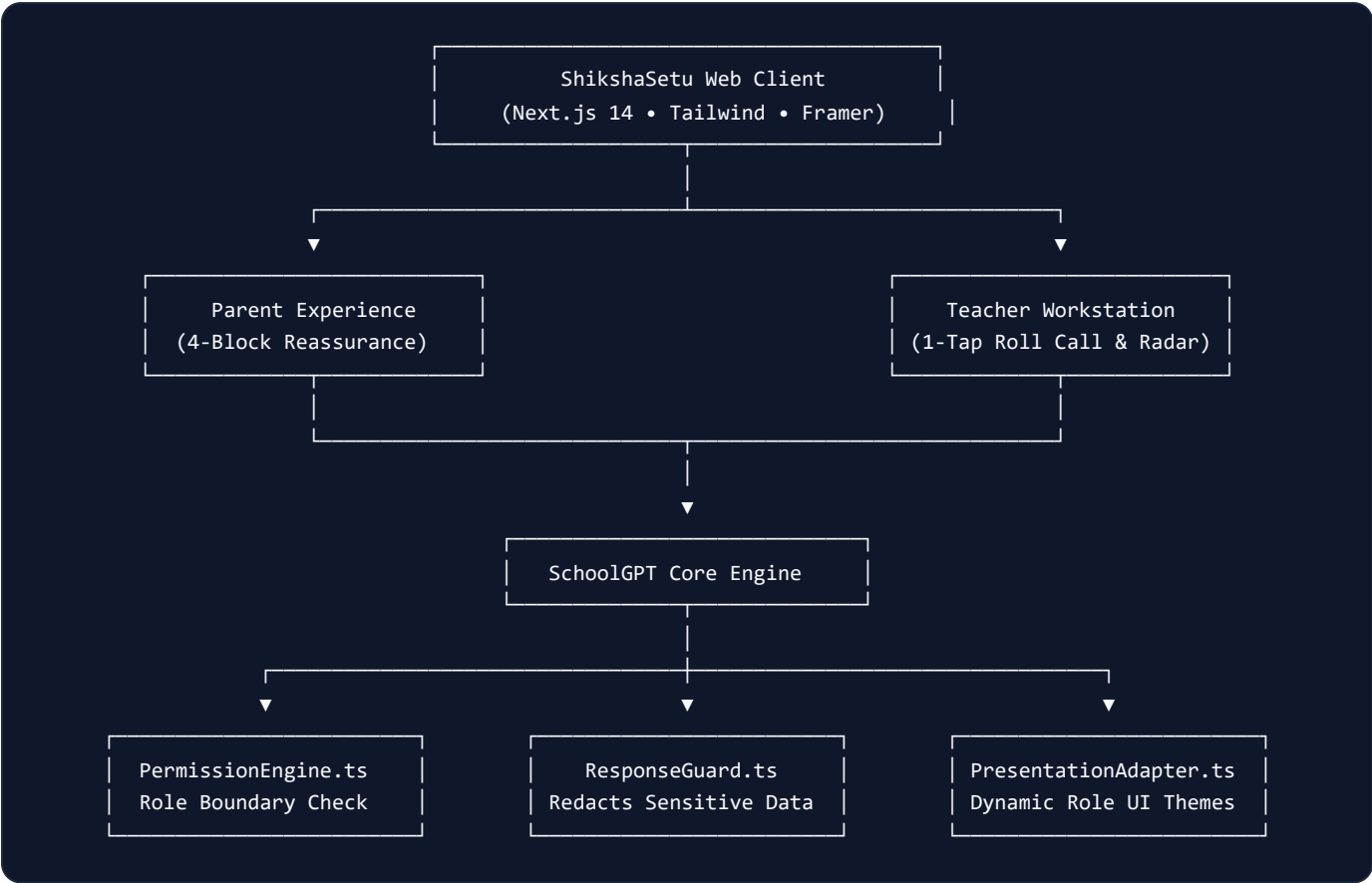
Connecting both portals is **SchoolGPT**—a unified intelligence engine equipped with role-aware context adapters and strict permission boundary enforcement.

2. Product Vision & Core Design Principles

Core Philosophy

- **Time to First Wow**: A hackathon judge or parent should understand the product's value in under 10 seconds and start experiencing the product in under 20 seconds.
- **Reassurance Over Metrics**: Avoid raw data dump. Tell parents directly whether their child is safe and on track today.
- **One Question Per Page**:
- **Home**: *Is everything okay today?*
- **Bus**: *Where is my child right now?*
- **Homework**: *What needs to be completed?*
- **Messages**: *Do I need to communicate with the school?*
- **Attendance**: *Is attendance healthy?*
- **Editorial & Handcrafted Design System**: Use sleek HSL gradients, dark modes, modern typography (`Outfit` / `Inter`), generous whitespace, and subtle micro-animations, avoiding generic SaaS template looks.

3. System Architecture



Technology Stack

- **Frontend:** Next.js 14 (App Router, Server Actions), TypeScript, TailwindCSS, Framer Motion.
- **Mapping & GIS:** Leaflet.js, OpenStreetMap, OSRM routing engine.
- **Backend & Database:** Supabase (PostgreSQL, Realtime Subscriptions, Row Level Security).
- **AI Intelligence:** Unified SchoolGPT Core with custom prompt composition, intent classification, and permission filtering.

4. Portal & Feature Implementation Breakdown

4.1 Landing Page (Dual Experience Hub)

- **Hero Section** (`components/landing/HeroSection.tsx`): Clear value proposition with primary CTA button and instant access pills (`Parent Experience` , `Teacher Experience`).
- **Dual Showcase Hub** (`components/landing/DualExperienceHubSection.tsx`): Split interactive container featuring the **Mobile Parent Companion** (left) and **Laptop Teacher Workstation** (right), visually connected by the central SchoolGPT Intelligence beam.
- **Performance Optimizations:** Optimized for hackathon judges—product experience accessible within 1 click.

4.2 Parent Experience (/parent)

Located in `components/parent/ParentTodayClient.tsx`, refactored into a focused 5-tab layout:

1. 4-Block Reassurance Home (`activeNav === 'home'`):

- *Block 1 — Child Status Card*: Clear confirmation ( Aarav arrived safely. Verified at Gate 2 • 08:18 AM).
- *Block 2 — Live Bus Tracking Snapshot*: Route ETA and telemetry (Next Stop: Saket Community Center • ETA 08:42 AM).
- *Block 3 — Important Updates Grid*: Algebra homework due, teacher notes, upcoming PTM.
- *Block 4 — SchoolGPT Assistant Bar*: 1-tap contextual queries.

2. Live Bus Tracking (`activeNav === 'bus'`):

- Interactive Leaflet map rendering real-time route path, driver details, and stop ETAs (`components/parent/ParentBusTrackingTab.tsx`).

3. Apple-Health-Style Attendance (`activeNav === 'attendance'`):

- *Level 1 — Reassurance Hero*:  Present Today (Verified at 08:18 AM), Monthly Attendance 90%.
- *Level 2 — SchoolGPT Insight*: Conversational summary ("Attendance has been consistently healthy this month. No action required.").
- *Level 3 — Monthly Calendar Grid*: Clean Apple Calendar heatmap with green/amber/rose indicators.
- *Level 4 — Recent History*: Truncated 3-log view with full history toggle.
- *Level 5 — Action Bar*:  Request Leave ,  Download Report ,  Contact Teacher .

4. 2-Column Teacher Workspace Messaging (`activeNav === 'messages'`):

- *Left Column (33%)*: Teacher profile (Ms. Ananya Mehra), Active status ( Online Now • Responds ~15 min), quick actions, and Today's Context snapshot.
- *Right Column (67%)*: Conversation timeline grouped by date,  **3 Suggested Replies**, multiline composer with attachment, translation, and  Improve with SchoolGPT tone refinement.

4.3 Teacher Workstation (/teacher)

Located in `components/teacher/TeacherDashboardClient.tsx`:

- **1-Tap Roll Call**: Batch marks all 40 students present in 1 click, with individual late/absent toggle overrides—saving teachers ~1.5 hours per week.
- **Student Support Radar**: Proactive identification of students needing academic or emotional guidance.
- **AI Lesson & Worksheet Generator**: Instant drafting of subject worksheets and PTM summaries.

4.4 Admin Mission Control (/admin)

Located in `components/admin/AdminDashboardClient.tsx`:

- Real-time campus attendance health telemetry, transport fleet speed alerts, gate entry logs, and fee collection overviews.

4.5 Driver & Gate Consoles (/driver , /gate)

- **Driver App:** Assigned route stop sequence, live passenger boarding manifest, emergency contact triggers.
- **Gate Security:** Sub-second RFID/QR digital gate pass verification, pickup authorization logs, visitor pass issuance.

5. SchoolGPT Architecture & Permission Boundary Engine

SchoolGPT is engineered as **ONE unified intelligence engine** that dynamically adapts to the active user's role while guaranteeing strict data isolation.



Core Architecture Modules (lib/schoolgpt/):

1. **PermissionEngine.ts** :
 - Defines matrix permissions (parent , teacher , student , admin , driver , gate , vendor).
 - `isQueryInRoleBoundary()` : Pre-LLM interception layer. Automatically catches out-of-scope prompts (e.g., student asking for teacher internal notes or salary ledgers) and returns polite boundary responses without wasting tokens.
2. **ResponseGuard.ts** :
 - Post-processing sanitizer that scans responses to redact sensitive fields (e.g., internal teacher risk scores, private financials).
3. **PresentationAdapter.ts** :
 - Converts domain context into role-personalized UI props:
 - **Student:** `Hi, Aarav 🧑🎓` (Theme: Indigo/Purple)
 - **Parent:** `Hi, Aarav's Parent 🧑👤` (Theme: Amber/Orange)
 - **Teacher:** `Hi, Ms. Mehra (Class 8A Teacher) 🧑🏫` (Theme: Teal/Emerald)
 - **Admin:** `Hi, Campus Administrator 🧑💼` (Theme: Slate/Navy)
 - **Driver:** `Hi, Transit Driver 🧑🚗` (Theme: Yellow/Dark)
 - **Gate:** `Hi, Gate Security 🧑🔒` (Theme: Emerald/Dark)
4. **SchoolGPTCore.ts** : Unified orchestrator connecting server actions in `app/actions/schoolgptActions.ts`.

6. UX Refactoring & "One Question Per Page" Mental Model

During optimization passes, we eliminated generic SaaS dashboard aesthetics:

Previous Problem	Refactored Solution
Artificial Safety Score percentages (98% Safe)	Replaced with explicit human statuses (✔ Verified Safe at Gate 2 • 08:18 AM)
Cryptographic Hashes on daily logs (#BUS4-PASS-OK)	Replaced with clean timestamped officer verifications (Verified by Security Officer Priya)
Overly dense Attendance KPIs	Refactored into Apple Health 5-level calm reassurance flow
Single-column basic message box	Transformed into 2-Column Slack/Apple Mail communication workspace
Hardcoded chatbot greetings (Hi Rahul)	Dynamic role-aware headers (Hi, Aarav's Parent 🧡)

7. Verification & System Health

- **TypeScript Compilation:** Executed `npx tsc --noEmit` — **0 compilation errors.**
- **Database & API Integration:** Verified Supabase server actions and state fallback handlers.
- **Responsive Layout:** Tested across mobile breakpoints, tablet frames, and desktop displays.

Documentation compiled locally for ShikshaSetu platform analysis.